

PYTHON PROGRAMMING ASSIGNMENT

Build Your Own Rule-Based Python Chatbot

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1. Aim

To build a simple rule-based conversational chatbot in Python that interacts continuously with the user, recognizes different messages, provides appropriate responses, and allows the user to exit the conversation.

2. Requirements Covered

1. Ask the user's name before the main conversation.
2. Use while True: to keep the conversation running.
3. Use input() to receive messages.
4. Clean user input using .lower().strip().
5. Recognize at least two greetings: hello and hi.
6. Answer at least three different questions or topics.
7. Use if / elif / else for decision making.
8. Use break when the user types bye or exit.
9. Use the user's name in at least one response.
10. Provide a useful response for unknown messages using else.
11. Use the in operator so variations such as 'hello bot' can trigger a greeting.

3. Algorithm / Logic

1. Start the chatbot.
2. Ask the user for their name.
3. Display a welcome message.
4. Start a continuous conversation using while True.
5. Read the user's message using input().
6. Convert the message to lowercase and remove extra spaces using .lower().strip().
7. Check for bye or exit and use break to end the conversation.
8. Check greetings such as hello and hi using the in operator.
9. Check other supported questions/topics and provide suitable responses.
10. If no rule matches, use else to provide a useful unknown-message response.
11. Repeat until the user exits.

4. Python Source Code

```
# Simple Rule-Based Python Chatbot
```

```
print("=====")
print("      WELCOME TO MY PYTHON CHATBOT      ")
print("=====")
```

```
name = input("What is your name? ").strip()
print(f"Nice to meet you, {name}!")
```

```

print("You can say hello, ask questions, or type 'bye' or 'exit' to stop.")
print("-" * 45)

while True:
    user_input = input(f"{name}: ").lower().strip()

    if user_input == "bye" or user_input == "exit":
        print(f"Chatbot: Goodbye, {name}! Have a great day!")
        break

    elif "hello" in user_input or "hi" in user_input:
        print(f"Chatbot: Hello {name}! How are you?")

    elif "your name" in user_input or "who are you" in user_input:
        print("Chatbot: I am a simple rule-based Python chatbot.")

    elif "python" in user_input:
        print("Chatbot: Python is a popular programming language that is easy to learn.")

    elif "help" in user_input:
        print("Chatbot: I can respond to greetings, answer questions about Python, "
              "and provide simple information.")

    elif "how are you" in user_input:
        print("Chatbot: I am doing great! Thanks for asking.")

    else:
        print("Chatbot: Sorry, I don't understand that yet.")
        print("Chatbot: Try saying hello, asking about Python, typing help, or saying bye.")

```

5. Sample Execution / Demonstration

The following screenshot shows the chatbot successfully running in Google Colab, including greeting recognition, question handling, unknown-input response, and exit handling.

```

=====
WELCOME TO MY PYTHON CHATBOT
=====
What is your name? Sujith

Nice to meet you, Sujith!
You can say hello, ask questions, or type 'bye' or 'exit' to stop.
-----
Sujith: Hello
Chatbot: Hello Sujith! How are you?
Sujith: Hello bot
Chatbot: Hello Sujith! How are you?
Sujith: what is python
Chatbot: Python is a popular programming language that is easy to learn.
Sujith: who are you
Chatbot: I am a simple rule-based Python chatbot.
Sujith: xyzabc
Chatbot: Sorry, I don't understand that yet.
Chatbot: Try saying hello, asking about Python, typing help, or saying bye.
Sujith: bye
Chatbot: Goodbye, Sujith! Have a great day!

```

Figure 1. Output of the rule-based Python chatbot executed in Google Colab.

6. Test Cases

Test Input	Expected Result
hello	Greeting response
HELLO	Greeting response after .lower()
hi	Greeting response

Test Input	Expected Result
hello bot	Greeting response using in operator
what is python	Python-related response
who are you	Chatbot identity response
unknown message	Useful response from else
bye / exit	Goodbye message and program terminates

7. Conclusion

The rule-based Python chatbot successfully follows the INPUT -> PROCESS -> RESPOND -> REPEAT -> EXIT flow. It accepts continuous user input, handles greetings and multiple topics, uses conditional logic, recognizes message variations, responds to unknown inputs, and exits when the user enters bye or exit.

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